Task 1 — Selection and Research

Jurisdiction-Aware AI Credit Decisioning Compliance Tool for American Express

1.1 Selected Regulated Entity: American Express

For this project, I chose American Express as my regulated entity. American Express gives me a more focused and manageable case. Although it is often known as a payment and card brand, its business is closely connected with card issuing, consumer credit, credit risk management, customer analytics, and payment services.

This makes American Express suitable for a jurisdiction-aware RegTech project. Credit card approval is not only a commercial decision, but also a regulated decision that affects consumers' access to credit. If AI or machine learning is used to support such decisions, the firm must consider not only prediction accuracy, but also explainability, fairness, model governance, and regulatory accountability.

American Express has a clear regulated presence in the United States. American Express National Bank is listed by the Office of the Comptroller of the Currency as a national bank, so the US part of this project is based on a real regulated banking entity rather than a hypothetical fintech platform (Office of the Comptroller of the Currency, n.d.). American Express also operates internationally. Its 2025 Annual Report refers to international partnerships and global card-related business activities, which supports the view that it operates across different markets and customer environments (American Express Company, 2026).

For the non-US jurisdiction, I chose the European Union. American Express has European operations, and American Express Europe S.A. is listed as a payment service provider with its statutory seat in Madrid and supervision by Banco de España (De Nederlandsche Bank, n.d.). The EU is also useful for this project because its approach to AI regulation is clearly different from the US approach. Under the EU AI Act, AI systems used to assess the creditworthiness of natural persons or establish credit scores are classified as high-risk AI systems, except where they are used for financial fraud detection (European Union, 2024).

Therefore, American Express provides a good case for comparing how the same AI-supported credit decision may create different compliance obligations in the United States and the European Union.

1.2 Selected Domain: AI Credit Scoring and Algorithmic Fairness

The domain I selected is AI-driven credit decisioning, with a focus on credit scoring, adverse action explainability, and algorithmic fairness monitoring.

In this project, I assume that American Express uses a machine learning model to support credit card application decisions. The model may use variables such as income, debt-to-income ratio, credit history length, missed payments, employment status, and other credit risk indicators to estimate whether an applicant should be approved, declined, or sent for manual review.

I chose this domain because it sits at the intersection of business value and public risk. From the firm's perspective, AI credit scoring can improve speed, consistency, and risk segmentation. It may help a card issuer process applications faster and identify higher-risk applicants more efficiently. However, from a consumer and regulatory perspective, the same system can create serious concerns. A model may reject an applicant without a meaningful explanation, rely on proxy variables that reproduce historical inequality, or generate different approval outcomes across groups even when protected attributes are not directly used.

This is why I see this domain as closely connected to the assignment theme. The brief emphasises that RegTech tools are shaped by different legislative and political choices across jurisdictions, not only by technical requirements. It also requires the tool to show how regulatory logic changes between the US and at least one non-US jurisdiction.

My project is therefore not intended to be a generic credit scoring model. Instead, I want to design a compliance layer around the model. The tool should ask: if the same AI credit decision is made in the US and in the EU, what different explanations, metrics, documentation, or human review processes should be triggered?

1.3 Regulatory Divergence: United States vs European Union

The key regulatory divergence in this project is shown below:

Area	United States	European Union
Main regulatory focus	Adverse action explainability and fair lending	High-risk AI governance and bias monitoring
Key question	Can the firm explain why this applicant was declined?	Is the AI credit system properly governed and monitored?
Main output needed	Specific and accurate adverse action reasons	Fairness metrics, governance checks, drift monitoring, human oversight
Tool design implication	Individual-level explanation is central	System-level monitoring is central

In the United States, the main design issue is whether the firm can explain an adverse credit decision clearly and specifically. The CFPB has stated that creditors using AI or complex algorithms must still provide specific and accurate reasons when taking adverse action against consumers (Consumer Financial Protection Bureau, 2023). For American Express, this means that the US mode of the tool should translate model outputs into understandable adverse action reasons, such as high debt-to-income ratio, short credit history, recent missed payments, or insufficient repayment record.

- The US mode therefore asks: Can American Express explain why this specific applicant was declined in a clear, accurate, and consumer-understandable way?

In the European Union, the concern is broader. Since creditworthiness-related AI systems are treated as high-risk under the EU AI Act, the tool should not only explain individual decisions, but also support model governance, fairness monitoring, documentation, and human oversight (European Union, 2024).

•The EU mode therefore asks: Is this AI credit decisioning system being governed and monitored properly as a high-risk AI system?

This difference is important. In the US, a declined application mainly triggers an explanation problem. In the EU, the same decision may also indicate a wider governance problem if the model shows biased approval rates, performance drift, or weak human oversight.

1.4 Why This Case Is Suitable for Tool Design

This case is suitable because the same model output can have different compliance meanings depending on the jurisdiction. For example, if the model rejects a credit card applicant, the US compliance team needs to know whether the rejection can be explained with specific adverse action reasons. In the EU, the same rejection should also be considered within a wider system-level review: whether the model produces unequal outcomes across groups, whether the data distribution has shifted, and whether human review is required.

This is why I describe my proposed tool as a jurisdiction-aware compliance layer, not just a dashboard. A normal dashboard may show approval rates and model scores. My tool should interpret those outputs differently depending on whether the active jurisdiction is the US or the EU.

The project is also feasible within the assignment scope. Since I do not have access to real American Express customer data, I will use synthetic credit application data, which the assignment allows when real data is difficult to obtain. The synthetic dataset can include applicant-level variables such as income, debt-to-income ratio, credit history length, missed payments, employment status, predicted approval probability, and final decision.

1.5 Proposed Tool Direction

The proposed tool will be called: FairCredit Jurisdiction Navigator

The tool will support American Express compliance, credit risk, and model governance teams when reviewing AI-supported credit card decisions across different jurisdictions. It will not replace the underlying credit model or make final approval decisions. Instead, it will sit above the model as a compliance and governance layer.

The tool will include: a jurisdiction configuration layer; synthetic applicant-level credit data; a machine learning credit approval model; individual-level adverse action reason generation; portfolio-level fairness and model drift monitoring; human review triggers; a model card or governance documentation stub.

The central design principle is:

The same AI credit decision should not be treated as having the same compliance meaning in every jurisdiction.

This principle keeps the project focused. In the US mode, the tool prioritises specific adverse action explanations. In the EU mode, it prioritises high-risk AI governance, fairness monitoring, drift detection, and human oversight. Overall, I chose American Express and AI credit decisioning because the case is specific enough to implement, but broad enough to show the real challenge of jurisdictional divergence in RegTech.

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